

Shock Absorber Simulation Validation Report

Topic: Modelling Energy Loss in a Shock Absorber

By: H068, H071, H097, H086

Title: Making Sure Our Shock Absorber Energy Loss Model Actually Works

1. Why We're Doing This

So, we built a simulation of a shock absorber to study how it loses energy—basically, how it turns motion into heat. But before we trust the results, we need to check if the physics inside the simulation makes sense. This report walks through how we validated that.

We did three things. First, we compared our simulation equations to standard textbook formulas. Second, we checked whether energy is conserved properly (or rather, whether the energy we lose to damping matches what we expect). Third, we tweaked key parameters like the damping coefficient and fluid viscosity to see if the simulation reacts in a realistic way.

Shock absorbers are tricky because you've got spring forces, damping forces, friction from seals, and even gas pressure fighting back. Our simulation tries to capture all that, but without going overboard on complexity.

2. How Our Equations Stack Up Against Textbooks

We looked at the main forces in our simulation and compared them to what you'd find in a mechanical vibrations or fluid mechanics textbook.

Damping force (linear model)

In our simulation: $F_d = -c \cdot v$

Textbook says: $F_d = c \cdot v$

The only difference is the negative sign—we just use that to make sure damping always pushes opposite to motion. Same physics.

Damping force (quadratic model for high speeds)

Our simulation: $F_d = -c_2 \cdot v \cdot |v|$

Textbook: $F_d = \frac{1}{2} \rho C_d A v^2$

They're really the same thing. The quadratic form is what happens when fluid flows through small holes in the piston at high speed—very common in real shock absorbers.

Spring force

Both us and the textbook: $F_s = -k \cdot x$

No surprises here. Some shock absorbers also have a gas spring, so we threw in $F_{gas} = \frac{P_0 A}{(1-x/x_{max})^\gamma}$ for those cases.

Seal and piston friction

Our simulation: $F_f = \mu \cdot F_N \cdot \text{sign}(v)$

Textbook: Same basic idea, though many books just treat it as simple Coulomb friction.

Bottom line: our equations match up nicely with theory.

3. Checking the Energy Balance

If our simulation is physically correct, the energy you put in should either get stored (in the spring) or turned into heat (by the damper). So we tracked everything.

Kinetic energy

$$T = \frac{1}{2}mv^2$$

Spring potential energy

$$V = \frac{1}{2}kx^2$$

Total mechanical energy

$$E_{mech} = T + V$$

Power lost to damping

$$P_{loss} = F_d \cdot v = cv^2$$

Total energy dissipated

$$E_{loss} = \int F_d \cdot v dt = \int cv^2 dt$$

We made sure that:

$$E_{input} \approx E_{stored} + E_{loss} + E_{friction}$$

There's always a tiny leftover error, but it's small enough to tell us the simulation is numerically stable. Over one full compression and rebound cycle, the mechanical energy we started with ends up almost exactly equal to the heat we dissipated. That's a good sign.

4. What Happens When We Change Things?

We wanted to see if the simulation responds the way a real shock absorber would, so we played with a few knobs.

Damping coefficient (c)

Crank it up, and the oscillation dies down way faster. But you also get bigger forces transmitted to the car body. Energy loss per cycle goes up linearly—makes sense.

Fluid viscosity

Thicker fluid means more damping and more heat. But if you go too far, you risk cavitation (air bubbles forming) or overheating. The simulation shows exactly that trend.

Orifice size

Make the little holes in the piston smaller, and the quadratic damping effect kicks in harder at high speeds. This is how real shock tuners adjust "high-speed compression" damping.

Gas pressure (for gas shocks)

Higher gas pressure helps prevent cavitation and adds a bit of spring force, but it doesn't change total energy loss over a full cycle by much. Simulation matches reality here too.

Overall, tweaking parameters gives you exactly what you'd expect. So the model passes the "does it feel right?" test.

5. What Works and What Doesn't

The good news: our simulation does a solid job of capturing how shock absorbers lose energy. Using both linear and quadratic damping lets us handle low-speed and high-speed behaviour pretty well.

The spring force, friction, and gas spring terms all work together to make the model behave like a real suspension component.

But let's be honest—there are some things we left out:

- Temperature changes: real shock fluid gets thinner as it heats up. We don't model that yet.
- Cavitation: when the fluid vaporizes at high speeds, damping drops off. Not in our model.
- Seal friction hysteresis: real seals don't behave exactly like simple Coulomb friction. They have memory effects.

Still, even with these limitations, the model is a solid starting point. Next steps could be adding thermal effects and validating against dyno test data.

6. Conclusion

We validated our shock absorber simulation by comparing it to textbook equations, checking energy conservation, and running sensitivity tests. Everything behaves the way physics says it should.

So yeah—we're confident this model can be used for teaching, suspension design, or just experimenting with how energy gets turned into heat in mechanical systems. It's not perfect, but it's physically sound.